

Detection of a Common and Persistent tet(L)-Carrying Plasmid in Chicken-Waste-Impacted Farm Soil



The connection between farm-generated animal waste and the dissemination of antibiotic resistance in soil microbial communities, via mobile genetic elements, remains obscure. In this study, electromagnetic induction (EMI) surveying of a broiler chicken farm assisted soil sampling from a chicken-waste-impacted site and a marginally affected site. Consistent with the EMI survey, a disparity existed between the two sites with regard to soil pH, tetracycline resistance (Tcr) levels among culturable soil bacteria, and the incidence and prevalence of several tet and erm genes in the soils. No significant difference was observed in these aspects between the marginally affected site and several sites in a relatively pristine regional forest. When the farm was in operation, tet(L), tet(M), tet(O), erm(A), erm(B), and erm(C) genes were detected in the waste-affected soil. Two years after all waste was removed from the farm, tet(L), tet(M), tet(O), and erm(C) genes were still detected. The abundances of tet(L), tet(O), and erm(B) were measured using quantitative PCR, and the copy numbers of each were normalized to eubacterial 16S rRNA gene copy numbers. tet(L) was the most prevalent gene, whereas tet(O) was the most persistent, although all declined over the 2-year period. A mobilizable plasmid carrying tet(L) was identified in seven of 14 Tcr soil isolates. The plasmid's hosts were identified as species of *Bhargavaea*, *Sporosarcina*, and *Bacillus*. The plasmid's mobilization (mob) gene was quantified to estimate its prevalence in the soil, and the ratio of tet(L) to mob was shown to have changed from 34:1 to 1:1 over the 2-year sampling period.